20-40插件框架单元测试

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1.单元测试框架工程介绍

2.编辑单元测试方法示例

2.1插件测试脚本示例

插件测试示例 展开源码

```
using System;
using System.Configuration;
using Microsoft.VisualStudio.TestTools.UnitTesting;
using Microsoft.Xrm.Tooling.Connector;
using Vyung.Xrm.Logging;
using Rhino.Mocks;
using Microsoft.Xrm.Sdk;
using Microsoft.Xrm.Sdk.Query;
using Vyung.Crm.Plugins;
using Vyung.Xrm.Plugin.Core;
using Vyung.Xrm.Config;
using System.Collections.Generic;
```

```
namespace Vyung.Crm.UnitTest
{
[TestClass]
public class PluginTest : ServiceProviderBase
{
/// <summary>
/// 执行其他环境
/// </summary>
public PluginTest() : base(EnvironmentType.DEV) { }

[TestMethod]
[Description("模拟测试Action")]
public void PluginTemplateTest1()
{
var paramBag = new Dictionary<string, object>();
paramBag.Add("inputPara1", "1000001");
paramBag.Add("inputPara2", "parameter");

var serviceProvider = ConstructServiceProvider(paramBag);
PluginTemplate test = new PluginTemplate("","");
test.Execute(serviceProvider);

Assert.AreEqual(this.output["state"].ToString(), "success", "执行失败");
}

[TestMethod]
[Description("模拟测试Plugin")]
public void PluginTemplateTest2()
{
/* 当前模拟更新后操作,界面执行提交操作new_status改成30,
* 插件已注册前后镜像 */
var paramBag = new Dictionary<string, object>();
var id = Guid.Parse("当前单据ID");

//模拟前镜像 后镜像
var preImage = serviceProxy.Retrieve("account", id, new ColumnSet(true));
var postImage = serviceProxy.Retrieve("account", id, new ColumnSet(true));
postImage["new_status"] = new OptionSetValue(30);
```

```
//模拟Target操作
var target = new Entity("account", id);
target["new_status"] = new OptionSetValue(30);
paramBag.Add("Target", target);

//模拟插件操作
base.RegisterSteps(MsgType.Update, 40);
base.RegisterPreImage(preImage);
base.RegisterPostImage(postImage);

//模拟当前操作人context.userid .
//不使用此方法默认当前代理用户权限.
base.RegisterUserId("操作人Guid");

var serviceProvider = ConstructServiceProvider(paramBag);
PluginTemplate test = new PluginTemplate("", "");
test.Execute(serviceProvider);
}
}
}
```

2.2控制台测试脚本示例

控制台任务测试脚本示例 展开源码

```
using System;
using System.Collections.Generic;
using System.Configuration;
using System.Data.SqlClient;
using System.Linq;
using System.ServiceModel;
using Microsoft.VisualStudio.TestTools.UnitTesting;
using Microsoft.Xrm.Sdk;
using Microsoft.Xrm.Tooling.Connector;
using Vyung.Crm.Consoles;
using Vyung.Xrm.Config;
using Vyung.Xrm.Console.Core;
using Vyung.Xrm.Logging;
```

```
namespace Vyung.Crm.UnitTest
{
[TestClass]
public class ConsoleTest
{
[TestMethod]
public void TaskTemplateTest()
{
//"TaskTemplate":任务名称, "para01"：参数值,"para02"：参数值
string[] args = new string[] { "TaskTemplate", "para01","para02"};
TaskContext taskContext = new TaskContext();
string taskCode = args[0];
try
{
List<string> cmdarg = new List<string>();
if (args.Length > 1)
{
cmdarg.AddRange(args.Skip<string>(1));
}
ConfigFactory config = new ConfigFactory(ConfigurationManager.AppSettings["xrmConfig"].ToString());
taskContext.Args = args;
taskContext.CrmService = new CrmServiceClient(config.GetCRMConnectionString());
taskContext.DbConnection = new SqlConnection(config.GetDBConnectionString());
//taskContext.Logger = new FileLogger(config.GetNLogPath());//配置文件Log
//执行任务
//taskContext.Logger.ConsoleInfo("{taskCode}程序启动时间：{DateTime.Now.ToString() }"); ITask taskName = TaskFactory.CreatTaks(taskCode);
taskName.Execute(taskContext); // taskContext.Logger.ConsoleInfo("{taskCode}程序结束时间：{DateTime.Now.ToString() }");
}
catch (FaultException<OrganizationServiceFault> ex)
{
// taskContext.Logger.ConsoleError($"{taskCode}程序运行失败：Message:{ex.Message}", ex);
throw;
}
}
}
}
```

3.配置文件使用

3.1 可配置框架读取配置文件的路径，路径可以调整。app.config：<add key="xrmConfig" value="C:\CRMConfig\Vyung.Xrm.config" />

App.config 展开源码

```
<configuration>
<appSettings>
<add key="xrmConfig" value="C:\CRMConfig\Vyung.Xrm.config" />
</appSettings>
<runtime>
<assemblyBinding xmlns="urn:schemas-microsoft-com:asm.v1">
<dependentAssembly>
<assemblyIdentity name="Newtonsoft.Json" publicKeyToken="30ad4fe6b2a6aeed" culture="neutral" />
<bindingRedirect oldVersion="0.0.0.0-10.0.0.0" newVersion="10.0.0.0" />
</dependentAssembly>
</assemblyBinding>
</runtime>
</configuration>
```

3.2配置框架文件Vyung.Xrm.config

增加crmConfig节点配置：

```
<!--start测试工程环境配置-->
<add key="CRMDEVConnectionString" value="Url=https://hostname.contoso.com; Domain=xxx; Username=xxx\xxx; Password=xxx; AuthType=IFD;" />
<add key="CRMUATConnectionString" value="Url=https://hostname.contoso.com; Domain=xxx; Username=xxx\xxx; Password=xxx; AuthType=IFD;" />
<add key="CRMPRDConnectionString" value="Url=https://hostname.contoso.com; Domain=xxx; Username=xxx\xxx; Password=xxx; AuthType=IFD;" />

<crmConfig>

<add key="CRMConnectionString" value="Url=http://contoso.vyung.com:100/orgname; Domain=xxx; Username=xxx; Password=xxx; AuthType=AD;" />
<!--start测试工程环境配置-->
<add key="CRMDEVConnectionString" value="Url=https://hostname.contoso.com; Domain=xxx; Username=xxx\xxx; Password=xxx\xxx; AuthType=IFD;"
/>
<add key="CRMUATConnectionString" value="Url=https://hostname.contoso.com; Domain=xxx; Username=xxx\xxx; Password=xxx; AuthType=IFD;" />
<add key="CRMPRDConnectionString" value="Url=https://hostname.contoso.com; Domain=xxx; Username=xxx\xxx; Password=xxx; AuthType=IFD;" />

<!--end测试工程环境配置-->
<!-- Online using Office365 -->
<!--<add key="CRMConnectionString" value="Url=https://contoso.crm.dynamics.com; Username=crmadmin@contoso.onmicrosoft.com;
Password=password; AuthType=Office365;" />-->

<!--<add key="CRMConnectionString" value="Url=https://hostname.contoso.com/orgname; Domain=mydomain; Username=mydomain\administrator;
Password=password; AuthType=IFD" />-->
```

```
</crmConfig>

<!--DB配置-->
<dbConfig>
<add key="DBConnectionString" value="Initial Catalog=demo;Data Source=.;UID=sa;Password=zhang123;" />
</dbConfig>
<!--消息配置-->
<messageConfig>
<!--短信消息配置-->
<msm>
<add key="sign" value="【xxxx】" />
<add key="account" value="dh10571" />
<add key="passwd" value="yksS9Z87" />
<add key="smsApiServer" value="http://xxxx.com/send" />
</msm>
<!--微信消息配置-->
<wechart>
<add key="appId" value="wxb10093447b230f3f"></add>
<add key="appSecret" value="58bddc219b9dc4bfa019c618b60a8e78"></add>
</wechart>
<!--邮件消息配置-->
<email>
<add key="emailServer" value="Smtp邮件服务器"></add>
<add key="from" value="发送邮件地址"></add>
<add key="userName" value="发送邮件账号"></add>
<add key="passWord" value="发送邮件密码"></add>
</email>
</messageConfig>

<!--日志配置-->
<nlogConfig>
<add key="nlogPath" value="C:\CRMConfig\NLog.config"></add>
</nlogConfig>

<!--扩展配置-->
<appConfig>
<add key="key01" value="001"></add>
```

```
<add key="key02" value="002"></add>
</appConfig>

</xrmConfig>
```

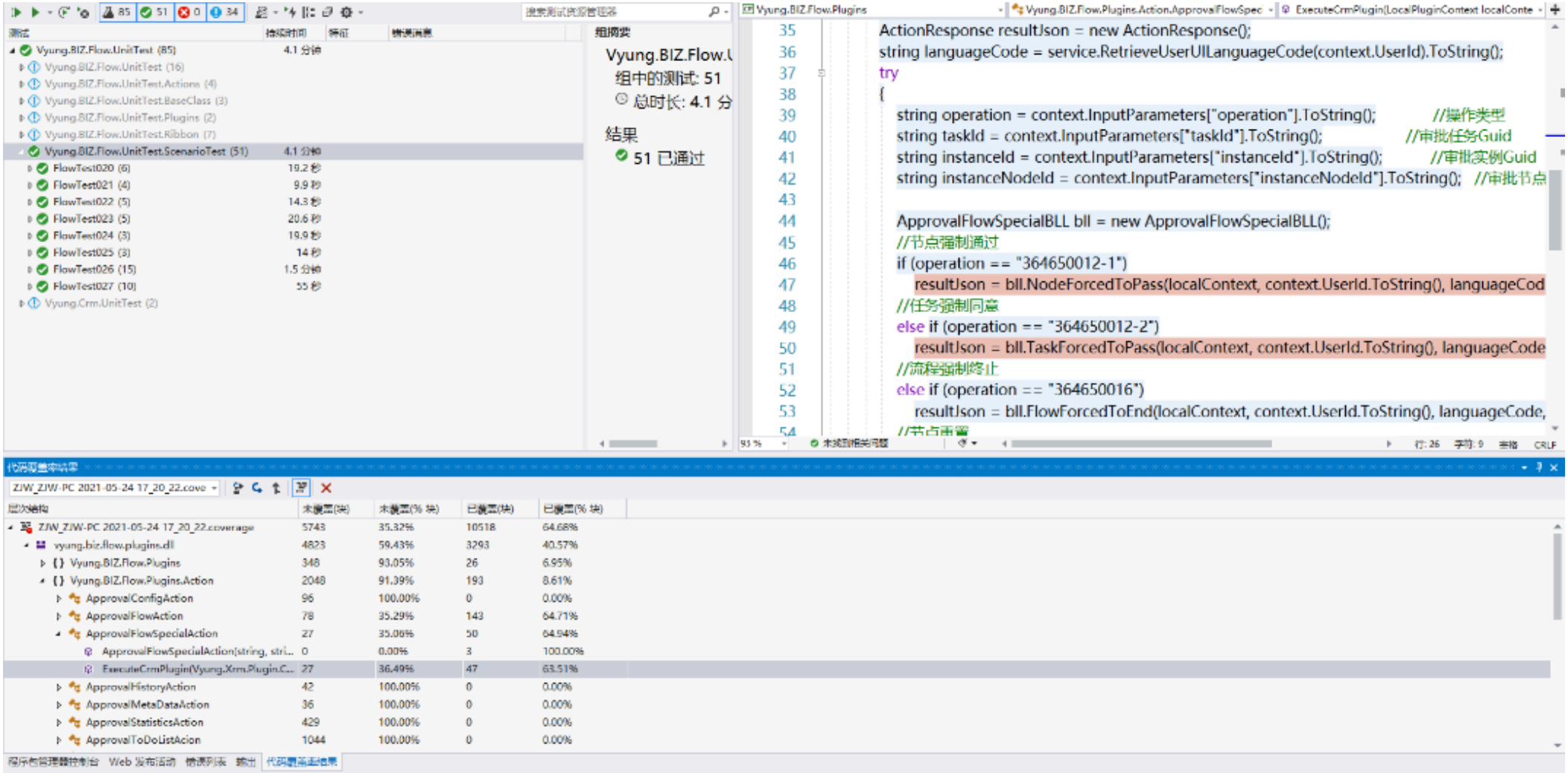
3.3配置框架文件NLog.config
可配置框架写日志的路径，路径可以调整
<variable name="BasicLogPath" value="C:\CRMConfig\Logs"/>
NLog.config 展开源码

```
<nlog xmlns="http://www.nlog-project.org/schemas/NLog.xsd" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.nlog-project.org/schemas/NLog.xsd NLog.xsd" autoReload="true">

<variable name="BasicLogPath" value="C:\CRMConfig\Logs"/> \<variable name="BasicLayout" value="${message}" /> <variable
name="PuginLogFileName" value="${BasicLogPath}/Plugin ${shortdate}.log" /> <variable name="ConsoleLogFileName"
value="${BasicLogPath}/Console ${shortdate}.log" /> <variable name="WebLogFileName" value="${BasicLogPath}/Web ${shortdate}.log" />
<targets>
<target name="PluginInfo" xsi:type="File" fileName="PuginLogFileName" layout = "{BasicLayout}" encoding="UTF-8" />
<target name="PluginError" xsi:type="File" fileName="PuginLogFileName" layout = "{BasicLayout}{newline}" encoding="UTF-8" /> \<target
name="ConsoleInfo" xsi:type="File" fileName="{ConsoleLogFileName}" layout="{BasicLayout}" encoding="UTF-8"/> \<target name="ConsoleError"
xsi:type="File" fileName="{ConsoleLogFileName}" layout="BasicLayout{newline}" encoding="UTF-8" />
<target name="WebInfo" xsi:type="File" fileName="WebLogFileName" layout = "{BasicLayout}" encoding="UTF-8"/>
<target name="WebError" xsi:type="File" fileName="WebLogFileName" layout = "{BasicLayout}${newline}" encoding="UTF-8"/>
</targets>

<rules>
<logger name="PluginLog" minlevel="Info" writeTo="PluginInfo" final="true" />
<logger name="PluginErrorLog" level="Error" writeTo="PluginError" final="true" />
<logger name="ConsoleLog" minlevel="Info" writeTo="ConsoleInfo" final="true" />
<logger name="ConsoleErrorLog" level="Error" writeTo="ConsoleError" final="true" />
<logger name="WebLog" minlevel="Info" writeTo="WebInfo" final="true" />
<logger name="WebErrorLog" level="Error" writeTo="WebError" final="true" />
</rules>
</nlog>
```

4.分析单元测试覆盖率


Vyung.BIZ.Flow.UnitTest (85) 4.1 分钟
Vyung.BIZ.Flow.UnitTest (16)
Vyung.BIZ.Flow.UnitTest.Actions (4)
Vyung.BIZ.Flow.UnitTest.BaseClass (3)
Vyung.BIZ.Flow.UnitTest.Plugins (2)
Vyung.BIZ.Flow.UnitTest.Ribbon (7)
Vyung.BIZ.Flow.UnitTest.ScenarioTest (51) 4.1 分钟
FlowTest020 (6) 19.2 秒
FlowTest021 (4) 9.9 秒
FlowTest022 (5) 14.3 秒
FlowTest023 (5) 20.6 秒
FlowTest024 (3) 19.9 秒
FlowTest025 (3) 14 秒
FlowTest026 (15) 1.5 分钟
FlowTest027 (10) 55 秒
Vyung.Crm.UnitTest (2)
组中的测试: 51
总时长: 4.1 分
结果
51 已通过
ActionResponse resultJson = new ActionResponse();
string languageCode = service.RetrieveUserUILanguageCode(context.UserId).ToString();
try
{
string operation = context.InputParameters["operation"].ToString(); //操作类型
string taskId = context.InputParameters["taskId"].ToString(); //审批任务Guid
string instanceId = context.InputParameters["instanceId"].ToString(); //审批实例Guid
string instanceNodeId = context.InputParameters["instanceNodeId"].ToString(); //审批节点
ApprovalFlowSpecialBLL bll = new ApprovalFlowSpecialBLL();
//节点强制通过
if (operation == "364650012-1")
resultJson = bll.NodeForcedToPass(localContext, context.UserId.ToString(), languageCod
//任务强制同意
else if (operation == "364650012-2")
resultJson = bll.TaskForcedToPass(localContext, context.UserId.ToString(), languageCode
//流程强制终止
else if (operation == "364650016")
resultJson = bll.FlowForcedToEnd(localContext, context.UserId.ToString(), languageCode,
ZJW_ZJW-PC 2021-05-24 17_20_22.coverage 5743 35.32% 10518 64.68%
vyung.biz.flow.plugins.dll 4823 59.43% 3293 40.57%
Vyung.BIZ.Flow.Plugins 348 93.05% 26 6.95%
Vyung.BIZ.Flow.Plugins.Action 2048 91.39% 193 8.61%
ApprovalConfigAction 96 100.00% 0 0.00%
ApprovalFlowAction 78 35.29% 143 64.71%
ApprovalFlowSpecialAction 27 35.06% 50 64.94%
ApprovalFlowSpecialAction(string, stri... 0 0.00% 3 100.00%
ExecuteCrmPlugin(Vyung.Xrm.Plugin.C... 27 36.49% 47 63.51%
ApprovalHistoryAction 42 100.00% 0 0.00%
ApprovalMetaDataAction 36 100.00% 0 0.00%
ApprovalStatisticsAction 429 100.00% 0 0.00%
ApprovalToDoListAction 1044 100.00% 0 0.00%